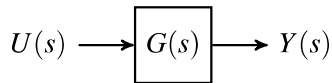


Chapter 2

Block Diagrams

Control systems generally consist of multiple devices (sensors, effectors) and a plant (electrical, mechanical, or a combination). These devices are linked together to form a system that performs a given function. The mathematical representation of each device is usually a transfer function. Devices are linked together by an input output relationship; where the output of one device becomes the input to another.

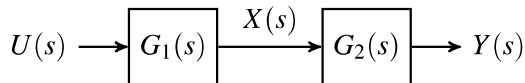
The most basic system consists of an input, output, and single transfer function



Mathematically, the input is $U(s)$ and an output is $Y(s)$. The output function $Y(s) = G(s)U(s)$, where $G(s)$ is a transfer function.

For simple problems cascaded as this example is, the algebra is simple. For more complex problems, a better representation is needed. The block diagram allows for transfer function representation and complex signal visualization without compromising accuracy.

The next example has an external input of $U(s)$ and an external output of $Y(s)$, however it is made up of two cascaded systems. The output signal of the first transfer function is the input the signal into the second transfer function. The signal between the first and second transfer functions (aka. blocks) is considered an **internal signal** while the input and output are **external signals**. When systems are cascaded together in such a way one can find the overall transfer function, $\frac{Y(s)}{U(s)}$.



To solve for the transfer function

$$Y(s) = G_2 X(s)$$

where

$$X(s) = G_1 U(s)$$

thus

$$Y(s) = G_2(s)(G_1 U(s))$$

or

$$Y(s) = G_1 G_2 U(s)$$

The transfer function is then

$$\frac{Y(s)}{U(s)} = G_1 G_2(s) \quad (2.1)$$

We use three fundamental constructs to develop complex representations of physical systems.

Transfer Functions

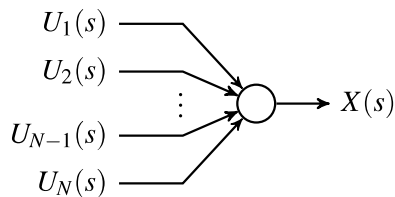
A linear, time-invariant, differential equation represented in the Laplace Domain as a ratio of output to input. to achieve a transfer function, the initial conditions must all be zero.

Summation Junctions

A function that sums all of the inputs into a single output

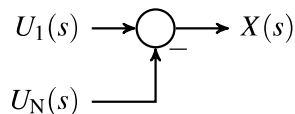
$$X(s) = \sum_{i=1}^N U_i(s) \quad (2.2)$$

where $U_i(s)$ are inputs and $X(s)$ is the output. The generalized summation diagram



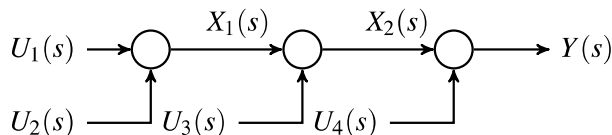
Additionally, the summation junction is used to add or subtract input signals. The sign operator is outside the summation circle near the signal to be subtracted (*Note: the absence of a sign designation implies a positive operator*).

Given the diagram

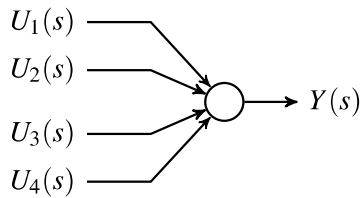


the output $X(s) = U_1(s) - U_N(s)$.

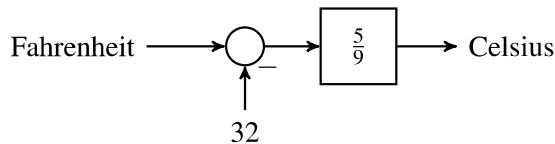
Summation junctions may be cascaded together, combined or separated. For example



has two internal signals, $X_1(s) = U_1(s) + U_2(s)$, $X_2(s) = X_1(s) + U_3(s)$, and the external signal $Y(s) = X_2(s) + U_4(s)$. The output may be represented as a summation of all of the external inputs, $Y(s) = U_1(s) + U_2(s) + U_3(s) + U_4(s)$ and as a diagram



Block diagrams can represent processes, both simple and complex. The diagram below uses a static gain and a summation junction to represent the mathematical operation to convert Fahrenheit to Celsius.

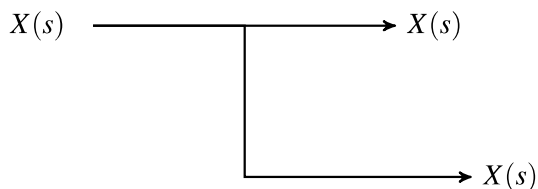


Which is

$$\text{Celsius} = \frac{5}{9}(\text{Fahrenheit} - 32)$$

Branches

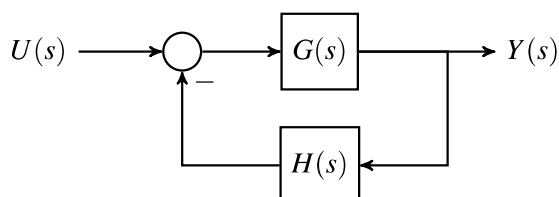
A branch is an ideal split such that the signal after the branch has two or more paths.



2.1 Manipulation

The block diagram approach is a good visualization tool that allows for quick combination of multiple devices by reducing the process to basic mathematics.

A typical model with negative feedback



where $G(s)$ is the forward path transfer function and $H(s)$ is the sensor feedback transfer function. Which produces an input/output transfer function

$$\frac{Y(s)}{U(s)} = T(s) = \frac{G(s)}{1 + GH(s)} \quad (2.3)$$

which can be written in **continued fraction** form by dividing the numerator and denominator by the numerator value

$$T(s) = \frac{G(s)}{1 + GH(s)} = \frac{\frac{G(s)}{G(s)}}{\frac{1 + GH(s)}{G(s)}} = \frac{1}{\frac{1}{G(s)} + H(s)} \quad (2.4)$$

The result in (2.1) is a useful tool for model construction.

For example, we are given a transfer function

$$T(s) = \frac{s + b_0}{s^2 + a_1s + a_0}$$

and we want to construct a block diagram with feedback. Simply find the continued fraction model of the transfer function and compare it to the general continued fraction model from (2.1).

$$T(s) = \frac{s + b_0}{s^2 + a_1s + a_0} = \frac{1}{\frac{s^2 + a_1s + a_0}{s + b_0}} = \frac{1}{\frac{s^2 + a_1s}{s + b_0} + \frac{a_0}{s + b_0}}$$

Now, compare the expression to the continued fraction model

$$T(s) = \frac{1}{\frac{s^2 + a_1s}{s + b_0} + \frac{a_0}{s + b_0}} = \frac{1}{\frac{1}{G(s)} + H(s)}$$

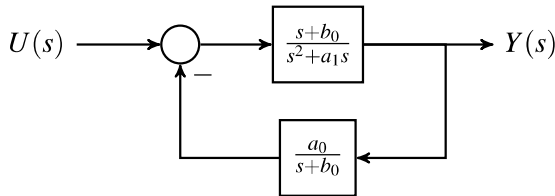
Now, one must make a choice as to the expressions for G and H . The rule of thumb: select the lowest order expression for H ¹. In this example

$$H(s) = \frac{a_0}{s + b_0}$$

and

$$\frac{1}{G(s)} = \frac{s^2 + a_1s}{s + b_0} \rightarrow G(s) = \frac{s + b_0}{s^2 + a_1s} = \frac{s + b_0}{s(s + a_1)}$$

Now, the block diagram for the transfer function $T(s)$ is

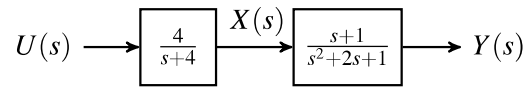


Let's consider some examples that demonstrate some of the most common block diagram manipulations

¹Selecting the wrong expression produces a model that will not work in software simulation tools

Example — Cascaded Blocks

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution:

$$Y(s) = \frac{s+1}{s^2+2s+1} \cdot X(s)$$

where

$$X(s) = \frac{4}{s+4} \cdot U(s)$$

$$Y(s) = \frac{s+1}{s^2+2s+1} \cdot \frac{4}{s+4} \cdot U(s) = \frac{4s+4}{s^3+6s^2+9s+4} U(s)$$

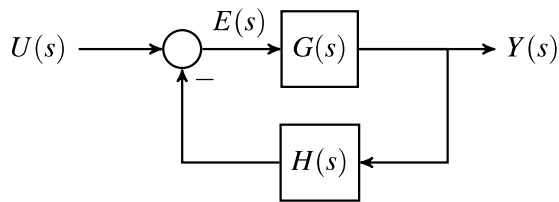
Finally

$$\frac{Y(s)}{U(s)} = \frac{4s+4}{s^3+6s^2+9s+4}$$

As is the custom, the final transfer function solution is written as a ratio of descending order polynomials.

Example — Negative Feedback

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution:

$$Y(s) = GE(s)$$

where

$$E(s) = U(s) - HY(s)$$

thus

$$Y(s) = G(s)(U(s) - HY(s))$$

$$Y(s) = GU(s) - GHY(s)$$

$$Y(s) + GHY(s) = GU(s)$$

$$(1 + GH(s))Y(s) = GU(s)$$

or

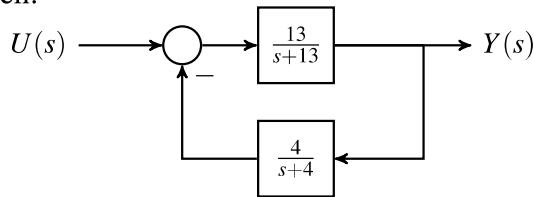
$$Y(s) = \frac{GU(s)}{1 + GH(s)}$$

Finally as a transfer function,

$$\frac{Y(s)}{U(s)} = \frac{G(s)}{1 + GH(s)} \quad (2.5)$$

Example — Another Negative Feedback

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution: The solution may be obtained using the block algebra from the previous example; here, let's solve it using direct mathematical analysis.

The output signal

$$Y(s) = \frac{13}{s+13} \left(U(s) - \frac{4}{s+4} Y(s) \right)$$

distribute the transfer function through the expression in the parentheses

$$Y(s) = \frac{13}{s+13} \cdot U(s) - \frac{52}{(s+13)(s+4)} Y(s)$$

collect the output signal Y on the lefthand side and factor out the signal

$$\left(1 + \frac{52}{(s+13)(s+4)} \right) Y(s) = \frac{13}{s+13} U(s)$$

write the expression in the parentheses as a fraction

$$\left(\frac{(s+13)(s+4) + 52}{(s+13)(s+4)} \right) Y(s) = \frac{13}{s+13} U(s)$$

isolate the output signal on the lefthand side

$$Y(s) = \left(\frac{(s+13)(s+4)}{(s+13)(s+4) + 52} \right) \frac{13}{s+13} U(s)$$

simplify

$$Y(s) = \left(\frac{13(s+4)}{(s+13)(s+4)+52} \right) U(s)$$

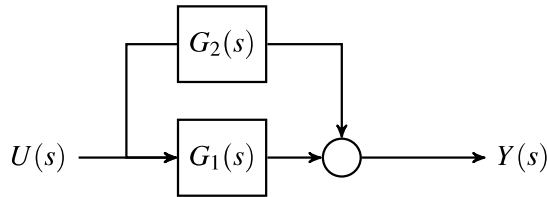
The final transfer function

$$Y(s) = \frac{13s+52}{s^2+17s+104} U(s)$$

Once again, the final transfer function solution is written as a ratio of descending order polynomials.

Example — Feedforward

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution:

$$Y(s) = G_1 U(s) + G_2 U(s)$$

thus,

$$Y(s) = (G_1(s) + G_2(s)) U(s)$$

Finally as a transfer function,

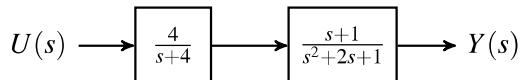
$$\frac{Y(s)}{U(s)} = G_1(s) + G_2(s) \quad (2.6)$$

2.2 Algebra

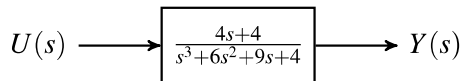
There are several algebraic approaches that allow us to manipulate block diagrams.

2.2.1 Cascaded Block Manipulation

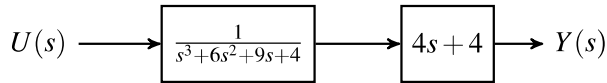
Just as above, we can manipulate cascaded blocks and form new math models. For example,



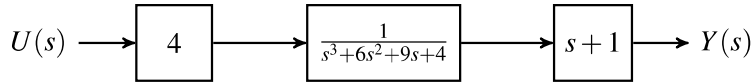
is the same as



which can be rewritten as



or as

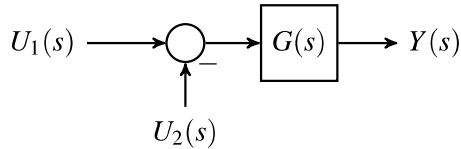


There are many potential representations – as long as the transfer function $\frac{Y(s)}{U(s)}$ is consistent among the models.

2.2.2 Moving Summation Junctions

After a Transfer function

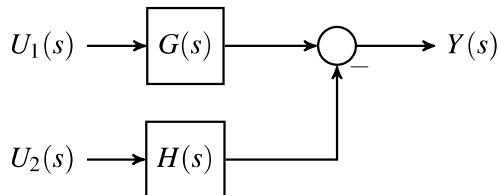
We have an original diagram



and we want to move the summation junction after the transfer function. We first consider the mathematical solution to the block diagram, which is

$$Y = G(s)(U_1(s) - U_2(s)) = GU_1(s) - GU_2(s) \quad (2.7)$$

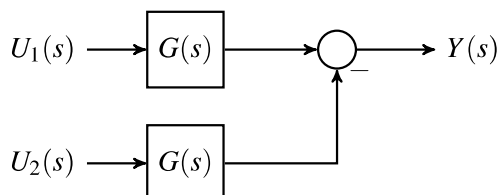
After moving the summation junction the solution **must** be the same



which produces an equation

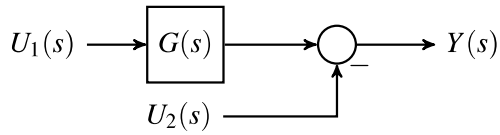
$$Y = GU_1(s) - HU_2(s) \quad (2.8)$$

Now, just compare (2.7) to (2.8) to resolve the unknown transfer function $H(s)$. Obviously, $H(s) = G(s)$, so the final block diagram is



Before a Transfer function

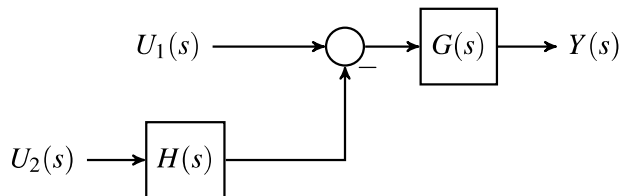
We have an original diagram



which produces an equation

$$Y(s) = GU_1(s) - U_2(s) \quad (2.9)$$

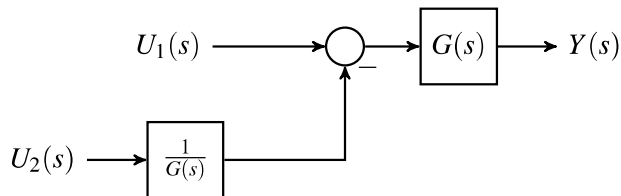
We want to move the summation junction before the transfer function to produce a block diagram



which produces the equation

$$Y(s) = G(s) \left(U_1(s) - HU_2(s) \right) = GU_1(s) - GHU_2(s) \quad (2.10)$$

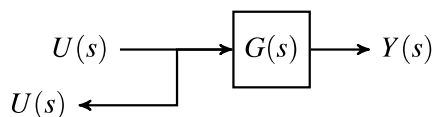
and now we just compare (2.9) to (2.10) to resolve the transfer function $H(s)$. If we set $H(s) = \frac{1}{G(s)}$ the equations will be equal and produce the final block diagram



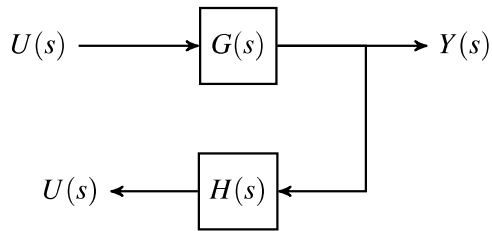
2.2.3 Moving Branches

After a Transfer function

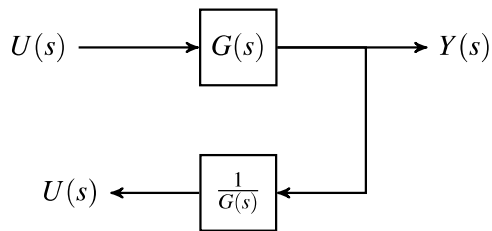
We start with a block diagram with an input signal that branches before going into a transfer function and we want to move the branch after the transfer function



Once again, after the move, we will have an unknown transfer function that we must determine

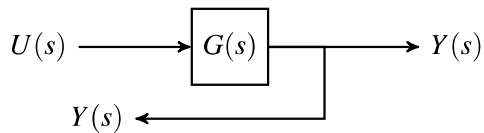


From the diagram $Y(s) = GU_1(s)$, therefore, $U_1(s) = \frac{1}{G(s)}Y(s)$, so the unknown transfer function $H(s) = \frac{1}{G(s)}$. The block diagram becomes

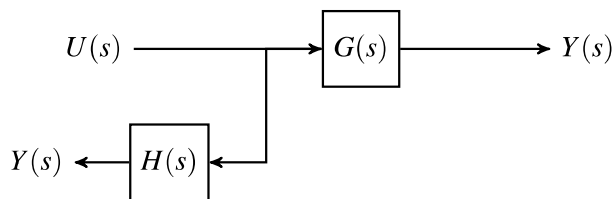


Before a Transfer function

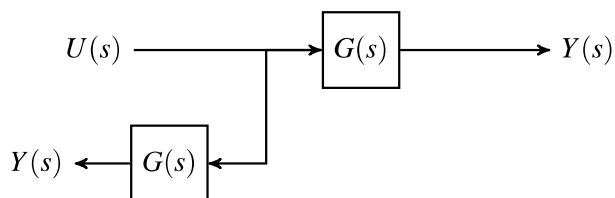
If we start with a block diagram with an output signal that branches after a transfer function and we want to move the branch before the transfer function



We repeat the process of placing an unknown transfer function in the block diagram

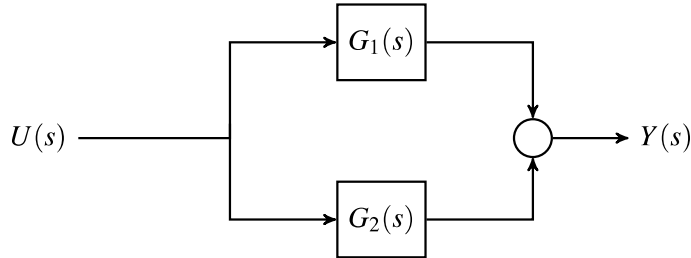


Of course, we know that $Y = GU(s)$, which means the unknown variable is simply $G(s)$. The block diagram becomes

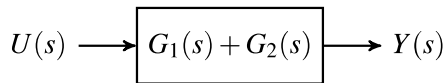


2.2.4 Combining Parallel Paths

When we have multiple transfer functions with the same input signal and their outputs are passed into a summation junction, we have parallel paths. See, Section 2.1 for an additional example.

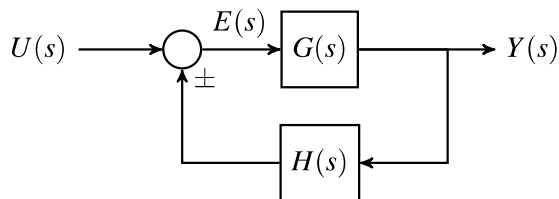


which produces an output signal $Y(s) = G_1U(s) + G_2U(s)$ and can be combined into a single transfer function



2.2.5 Eliminating a Feedback Loop

Section 2.1 is a good example of this concept, but let's formalize the process. We have a diagram with feedback (either positive or negative)



The output

$$Y(s) = G(s)E(s)$$

where

$$E(s) = U(s) \pm HY(s)$$

thus,

$$Y(s) = G(s)(U(s) \pm HY(s))$$

$$Y(s) = GU(s) \pm GHY(s)$$

$$Y(s) \mp GHY(s) = GU(s)$$

$$(1 \mp GH(s))Y(s) = GU(s)$$

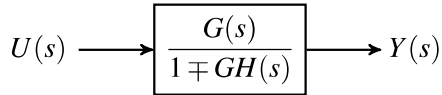
or

$$Y(s) = \frac{GU(s)}{1 \mp GH(s)}$$

which produces a transfer function

$$\frac{Y(s)}{U(s)} = \frac{G(s)}{1 \mp GH(s)}$$

and the block diagram simplifies to

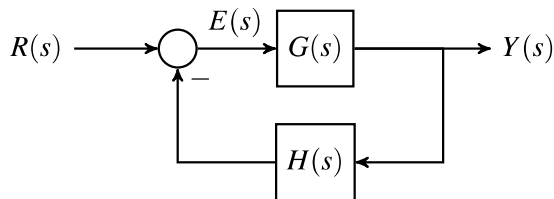


Which brings up an interesting concept and a common “conflict” in the field.

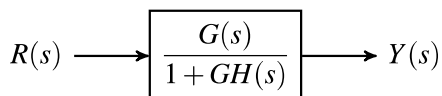
With this algebraic approach, we start with a diagram that has a feedback loop, and we simplify it to a diagram without a feedback loop. Additionally, we can reverse the process and start with a transfer function and end up with a diagram with a feedback loop.

Open-Loop vs Closed-Loop

Just like before we have a feedback system (I’m going to set the feedback to be negative to clean up the math, but the same concept applies for positive feedback as well).



which simplifies to



Visually, the first diagram has a feedback path (or loop), while the second does not. If the system has a loop then the signal can loop back on itself so it is closed-loop. Additionally, with the closed-loop system we define a **loop-gain** transfer function which is the product of all transfer functions within the loop; in this example $L(s) = GH(s)$.

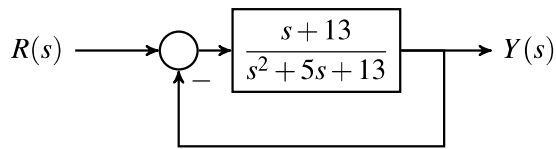
The second diagram does not loop back on itself, the signal passes from the input through the transfer to the output, so it is open-loop. I know, it should really be a “no loop system,” but that’s not the industry nomenclature – sorry.

What you won’t find in many source texts is presented in **Section 2.1**, which is the interoperability between the two models. For most systems, we can transform between the two models; each model presents unique analytical insights. The open-loop transfer function $\left(\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}\right)$ is written as a ratio of polynomials where the denominator $A(s)$ is the **characteristic polynomial** and determines stability.

The closed-loop transfer function $\left(\frac{Y(s)}{R(s)} = \frac{G(s)}{1+L(s)}\right)$ yields the loop-gain $L(s)$ which determines the response error and frequency stability – among other things. Often, we need to consider both models when analyzing a system or designing a controller.

Example

Given:



Find: The open-loop and closed-loop representations

By observation, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$ where

$$G(s) = \frac{s+13}{s^2+5s+13}$$

and

$$H(s) = 1$$

So

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)} = \frac{\frac{s+13}{s^2+5s+13}}{1 + \frac{s+13}{s^2+5s+13}}$$

which is the closed-loop model. Now, to find the open-loop, just simplify the closed-loop to a ratio of polynomials.

Write the closed-loop model as a ratio of polynomials to achieve the open-loop representation

$$\frac{Y(s)}{R(s)} = \frac{\frac{s+13}{s^2+5s+13}}{\frac{(s^2+5s+13)+s+13}{s^2+5s+13}} = \frac{s+13}{s^2+5s+13} \cdot \frac{s^2+5s+13}{s^2+6s+26} = \frac{s+13}{s^2+6s+26}$$

Now, we have both representations (**closed-loop** and **open-loop**)

$$\frac{\frac{s+13}{s^2+5s+13}}{1 + \frac{s+13}{s^2+5s+13}} \Leftrightarrow \frac{s+13}{s^2+6s+26}$$

which are equivalent, but offer different analytical insights into the system and how it will respond to various inputs.